QA-CANDIDATE-KNOWN-DEFECTS

QA candidate: known defects

Part 1 lists problems found while CONSTRUCTING this candidate (merge decisions, merge-induced breakage and the state of the quality gates run on the merged tree). Part 2 is the digest of the independent review reports for the merged streams. Nothing in Part 2 was re-run or re-reviewed for this build: statuses were checked by reading the code and commit log only.

Part 1: found while building the candidate

ID	Severity	Finding	Where
CC-01	HIGH	`verify-suppressed-residue.sh` exits 1 on the merged tree: ratchet FAIL for two NEW carriers of suppressed passage ids, `docs/faq.docx` (33 ids) and `docs/on-screen-text.md` (1), plus the known G3 debt. Both files arrive with the s11-018-library branch. These are served from `/docs`, so withheld text may be reachable in them.	candidate worktree `docs/`
CC-02	HIGH	`verify-redaction-propagation.sh` exits 1 (6 assertions failed; served `redactions` table 1156 rows against 1112 effective suppressions). Not diagnosed: the gate was run against the worktree, which lacks the untracked curriculum data.	gate output
CC-03	MED		

ID	Severity	Finding	Where
		The on-demand material path never sets the per-request acknowledgement (`WithRemoteAck`, only `/api/ask` does) and nothing in production code installs a spend hook (`SetSpendHook`), which the billable probe added by h1 now requires. A hosted provider row therefore cannot generate. The test `TestLazyHostedBuildsOnlyOnFirstGeneration` needed both added to pass (qa-fix). Hosted providers are OFF in this build, so this is not reachable.	`pkg/answer/hosted.go`, `cmd/workshop-server/wire_materialgen.go`
CC-04	LOW	h1 health probe runs BEFORE n3's per-request acknowledgement check inside `HostedProvider.Generate` (the probe carries no private text but is a billable call). Semantic merge choice, never reviewed.	`pkg/answer/hosted.go`
CC-05	MED	`pipeline/ocr/` holds two implementations: n6's spec-002 modules and n7's spec-008 modules share file names; the conflicted `__init__`, `corroboration`, `stability` take n7's. n6's `test_ocr.py` still passed 32/32 against the mixed package, but the package is not a reviewed design. Offline pipeline only, not served.	`pipeline/ocr/`
CC-06	LOW	The ask path with the `askStream` UX flag ON does not carry the hosted-send acknowledgement (`remoteAck`); the non-stream path does. Flags default OFF.	`features/inquiry/inquiry.component.ts`
CC-07	MED	`features/practice/adaptive-practice.component.ts` is n8's version; wire-completion's variant (announce / focus on a new question) was dropped. Its spec file is a union of both streams' expectations; Karma result for the merged tree is in the QA-CANDIDATE page, section 7 status line.	frontend
CC-08	LOW	`docs/faq.md` is the union of several streams: several `## 19.` / `## 20.` headings coexist. The generated `docs/*.pdf` / `*.html` exports of faq, manual and user-guide were NOT regenerated (they show one side only).	`docs/`

ID	Severity	Finding	Where
CC-09	LOW	n6's and n7's differently shaped `verify-gofmt.sh` gates share a file name; n6's ratchet is kept, n7's registry row `gofmt-clean` removed. The ratchet baseline was emptied (n7 had formatted the 13 files) and 3 files from other streams were formatted. Gate green after.	`platform/gates/`
CC-10	MED	My own conflict union of `platform/compose.yml` (n8) re-introduced the name-only `ApiKey_OpenAI` / `ApiKey_Anthropic` environment entries that the apikeys design removes (they put key values on the container command line once opted in). Caught by review of the digest, removed in a qa-fix; `scripts/api-keys.test.sh` 103 passed / 0 failed after.	`platform/compose.yml`
CC-11	LOW	The three-way union left research docs listed in the Library under the `[stale]` live-registry statements of some docs (e.g. "Not searchable yet" describes the un-ingested live registry; on the QA copy they ARE searchable).	docs
CC-12	INFO	Gates that need untracked data (`docs/research`, `platform/prebuilt`, `curriculum` build products) return UNDETERMINED (rc 2) when run from the worktree: `verify-research-reingest`, `verify-prebuilt-artifacts`, `verify-media-exposure`, `verify-server-unity` (5 PASS, 2 UNDET). They were not run against the QA copy.	gate output

Part 1 gates that PASSED on the merged tree (rc 0, run in this session): placeholder-sweep (source half), no-runtime-model-calls, medium-no-model, seed-password-absent, spec007-auth-wiring, scoring (200 tests), registry-content-type, limits-completeness, ui-labels, ui-vocabulary, no-hierarchy-fields, gofmt, source-ref-exposure, transcript-anchors, material-alt, material-metadata-disclosure, compose-provider-passthrough, `scripts/api-keys.test.sh`. | CC-13 | HIGH | Cross-stream: the roles stream (s18) is deny-by-default for student mutations but did not classify routes of other streams; on the merged tree students got 403 on practice answers/submit and on medium/materials mutations. Found by the merged Go suite, fixed on the candidate

(student allow-list + classification table, scanner now sees `reg(...)` registrations); s18's newer tip adds the practice routes itself, medium/materials remain candidate additions. Needs a real fix on the s18 branch. | `pkg/authhttp/roles.go` | FIXED on candidate only | | CC-14 | LOW | The `Study material reachability` Karma spec stubs AuthService; s18's nav now reads `isAdmin`/`roleLabel`, so the stub needed them (qa-fix). | frontend spec | FIXED on candidate | | CC-15 | LOW | Karma cannot start Chromium here without `--no-sandbox` (Ubuntu user-namespace restriction); a temporary karma config was used and not committed. | host | NOTE |

Part 2: digest of the independent reviews

Generated 2026-10-22 from the independent review reports against the candidate tree ~/Projects/vasic-wt/qa-candidate (branch qa-candidate, tip 7dff042). Read-only: no repository file was edited and no server was started. Statuses were checked with `git grep`, `git show` and `git log` on the candidate; where a later commit is cited its message was read, but no fix was re-run or re-reviewed, so LIKELY-FIXED-LATER is not a PASS.

Summary

- 57 rows in the main table. By severity: HIGH 9, MED 39, LOW 9. By status: OPEN 19, LIKELY-FIXED-LATER 27, DORMANT 8, UNKNOWN 3.
- Severity is normalised to HIGH / MED / LOW (BLOCKING and HIGH count as HIGH; MEDIUM-HIGH counts as MED, LOW-MEDIUM as LOW). LOW findings that no tester could see in the UI or API (test strength, evidence hygiene, parser differentials) are not listed.
- Appendix A lists 7 further rows against repositories that are NOT in the candidate (constitution governor, umbrella guard); they are not in the counts above.
- NO-GO or dormant per the reports: s9-014-modal-polish NO-GO as reviewed (fixed by 45c1ba2, fb6b5b7); s12-019-promotion NO-GO round 3 (round 4 commit 24c2200 merged, not re-reviewed); constitution governor-279 NO-GO and umbrella spec 027 (outside the candidate); s17-022-material is GO only with the feature OFF and is DORMANT (never attached, no route, no compose flag); s15-025-ux features are all behind flags that default to false; answering (spec 024) is OFF by default and hosted providers are forced off; spec 021 engagement is OFF by default.

- Reviews exist for n1, n2, n3 (twice), n5, wire-completion, s9, s12 (round 3), s16, s17 (round 4), h1, inv-nonascii, s2. No review report exists for n4, n6, n7, n8, s11, s15, s18, cat-authors; their merged tips are unreviewed here.
- Five findings most likely to hit a manual tester: (1) KD-026 provider key names are back in compose `environment:` (found by this digest in merge e96ff43; visible in `ps`/`podman inspect` once keys are opted in); (2) the Anthropic hosted row cannot start because no production code installs a spend hook; (3) KD-001 practice/progress slowdown on a hot question, with KD-002 self-rated flashcards inflating ability and level-ups and KD-003 the 'Noted: questions repeated after a miss' line on Progress for honest learners; (4) 019 deploy pieces not visible or gated (status panel `<wk-live-facts>` not mounted, `restart.sh` has no verdict gate); (5) the 025 UX wave and 022 materials are invisible with default settings (flags off, not attached), and 012 OCR through the ingest script fails (KD-005).

Main table

A. Adaptive practice and scoring (spec 013)

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
KD-001	013 US3 recalibration / 2PL fit	n1-013-adaptive (c7f9564)	HIGH	With one popular question answered by thousands of distinct sessions, the hourly difficulty recalibration and 2PL fit hold the single DB connection for seconds to minutes (n=4000: 9 s and 21 s; n=8000: 35 s and 84 s) and every practice or progress request waits. A tester sees practice/progress stall only at scale.	n1-2026-10-21.md F1
KD-002			MED		

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
	013 US2 flashcard easing	n1-013-adaptive (c7f9564)		Self-rated flashcards can raise ability without bound: a learner who answers every multiple-choice question wrong but taps 'recalled' on flashcards gains about +2 points (level-ups) because answered cards are served again. Reviewer rated it MEDIUM-HIGH.	n1-2026-10-21.md F2
KD-003	013 integrity reading / Progress page	n1-013-adaptive (c7f9564)	MED	An honest learner who finishes the question pool is flagged `repeat_after_wrong` (after 8/14/24/44 attempts for pools of 5/10/20/40), and the Progress page then prints 'Noted: questions repeated after a miss.' The flag counts the server's own stated repeats. Score is not changed.	n1-2026-10-21.md F3
KD-004	013 tests	n1-013-adaptive (c7f9564)	MED	Three scoring guards survive mutation (recalibration rapid-guess skip, `correct = true` for flashcards, FitIRT first-attempt filter), so a regression	n1-2026-10-21.md F4

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
				there would not turn tests red. Not visible to a tester.	

B. Textbook ingestion (spec 012)

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
KD-005	012 OCR	n2-012-ingest-complete (b3b0b1e)	HIGH	`--ocr` cannot run through `scripts/ingest-textbooks.sh`: the zero-token guard counts the 2 connect() calls made by `tesseract --version` and aborts (`ZERO-TOKEN GUARD TRIPPED`, extract rc 97, script rc 1). Scanned PDFs cannot be ingested by the documented command. OCR is off by default.	n2-2026-10-22.md F1
KD-006	012 HTML extractor	n2-012-ingest-complete (b3b0b1e)	MED	A valid HTML book with more than 200 unclosed `<p>`/`<li>`/`<td>` is rejected whole ('malformed HTML (nesting too deep)', 'no ingestable section', rc 1).	n2-2026-10-22.md F2
KD-007	012 HTML extractor	n2-012-ingest-complete (b3b0b1e)	MED	An HTML file with `<meta charset="hex">` (or base64/rot13/zlib) crashes the extractor with a LookupError traceback, rc 1 instead of the documented 'unreadable' rc 2.	n2-2026-10-22.md F3
KD-008			MED		

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
	012/017 docs and compose wording	n2-012-ingest-complete (b3b0b1e)		compose.yml, the rerank gate header and docs/cross-encoder-rerank.md say the cross-encoder defaults OFF 'on a measurement' (top-5 11/22 vs 15/22), but that number measured the in-window bm25 rerank, not the cross-encoder. Misleading text in operator-facing config and docs.	n2-2026-10-22.md F4
KD-009	012 OCR reporting	n2-012-ingest-complete (b3b0b1e)	MED	When every OCR page is below `--ocr-min-conf`, the message says 'every page below --min-chars' and no report is written, so the promised per-page confidence list never reaches the operator.	n2-2026-10-22.md F5
KD-010	012 DOCX/ EPUB/FB2 extractors	n2-012-ingest-complete (b3b0b1e)	MED	Silent content loss reported as `extraction_failures: []`: DOCX inline text boxes dropped; EPUB nested-table content duplicated and `<caption>` dropped; FB2 text placed directly in `<section>` dropped.	n2-2026-10-22.md F6

C. Answering gate, hosted spend control, figures (specs 020, 024)

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	S c
KD-011	024 L4b evidence-demand layer	n3-020-024-complete (bf2232e)	MED	Legitimate quantity answers were declined ('16GB', '512MB', '3rd', 'There are 3:'); re-review: shorthand '30s', '24h', '30d', '512Mi', '5min' read as no number, 40% of the reviewer's 95 legitimate answers declined. Safe direction (a decline), but a tester sees 'cannot answer'.	n3-2026-10-21.md #1; n3-rereview-2026-10-22.md F1	
KD-012	024 L4b evidence-demand layer	n3-020-024-complete (bf2232e)	MED	Fabrications admitted: the claim's number was never compared with the cited span's number or	n3-2026-10-21.md #2; n3-rereview-2026-10-22.md F2, F3, F4	

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	S c
				unit (999 MB vs 512 MB; 512 kilobytes vs 512 megabytes; euros vs dollars; monthly price presented as annual, because the period word only had to be within 8 tokens). Re-review: 34 of 50 fabrications admitted.		u a a g r 5 f s r t w a a r r
KD-013	024 spend ledger	n3-020-024-complete (bf2232e)	MED	'Corrupt ledger is never reset' had holes: a ledger of `{}`, `null` or without a schema key was accepted as empty and rewritten, restoring the cap after 9.5 USD spent; later in-place edit of `spent_usd` (and	n3-2026-10-21.md #3; n3-rereview-2026-10-22.md F9	I (' c (r g s r c j c r I S I t I c

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	S c
KD-014	020 figures route	n3-020-024-complete (bf2232e)	LOW	negative values) also undetected. The figure image route did not verify sha256/ bytes at serve time and the check stayed optional if the catalogue lacked the fields. A tester viewing / textbooks figures could be served an altered PNG.	n3-2026-10-21.md #7; n3-rereview-2026-10-22.md F13	

D. Toasts, badges, modal, profile (specs 014, 015)

ID	Spec / area	Branch (stream)	Severity	Finding	Source repo
KD-015	014 toasts and badges	wire-completion (ce9af74) + n5 (18d9eda)	HIGH	Toasts never fire against the real server: the HTTP event source omitted `events_after` on the first fetch, so the server returned no `achievement_events`	n5-2026-10-2 finding 1

ID	Spec / area	Branch (stream)	Severity	Finding	Source repo
				and the feed never primed.	
KD-016	014 badges	n5-011-014-015-complete (18d9eda)	MED	The badge producer re-walked the whole event history on every forced refresh (4 GETs per lesson mark; 12 for three rapid marks) and areas with more than 1000 events lost their earned-at stamp.	n5-2026-10-2 finding 2
KD-017	014 badges	n5-011-014-015-complete (18d9eda)	MED	Stale badge after un-marking the last touched lesson: /api/progress answers 404 and the producer treated 'empty' as an outage, so the badge stayed.	n5-2026-10-2 finding 3
KD-018	015 profile / document reader	n5-011-014-015-complete (18d9eda)	LOW	Document reader showed stale slug/location when reused between two / documents URLs and only breadcrumbs while the citation loaded; profile rendered the server's `last_downgrade_reason` verbatim.	n5-2026-10-2 findings 5 a
KD-019	014 modal (US2)	s9-014-modal-polish (2fef67b)	HIGH	Merging s9 into int turned Karma red (11 failures, storage-hygiene guard: modal ack and cursor keys left in storage). NO-GO as reviewed.	pending-branches-20 section 2 fin
KD-020			MED		

ID	Spec / area	Branch (stream)	Severity	Finding	Source repo
	014 modal evidence	s9-014-modal-polish (2fef67b)		Round-2 evidence (1051 SUCCESS, 50 e2e) was not backed by raw logs in the branch (round-1 logs still said 878).	pending-branches-20 section 2 fin
KD-021	014 modal queue	s9-014-modal-polish (2fef67b)	LOW	A pending modal beyond the 20-entry cap is silently dropped, and a queued-but-unseen modal is lost on tab close; the drop count was surfaced nowhere. Events earned between sign-in and the first prime can lose a level-up modal.	pending-branches-20 section 2 fin 4

E. Wiring of practice, providers, answering (wire-completion)

ID	Spec / area	Branch (stream)	Severity	Finding	Source repor
KD-022	011/024 FAQ and benchmark	wire-completion (ce9af74)	MED	FAQ 23 gave a 3-variable recipe to enable a LAN provider without saying that, without a matching gate verdict, it stays forced OFF; the default benchmark file questions-v2.tsv was missing from the tree.	wire-completion-20 F2
KD-023	013 adaptive practice UI	wire-completion (ce9af74)	LOW	On a refused serving token the card showed 'could not load (serving_token_invalid)' and dropped the	wire-completion-20 F3

ID	Spec / area	Branch (stream)	Severity	Finding	Source repor
				server's message ('nothing was scored, try again').	
KD-024	013 adaptive practice UI	wire-completion (ce9af74)	LOW	No focus move or announcement when a new question replaces the card (screen-reader users).	wire-completion-20 F4
KD-025	024 answering-off banner	wire-completion (ce9af74)	LOW	The banner said 'Searching and browsing are not affected' for any unavailable code, including a dead backend (http_0).	wire-completion-20 F5

F. Provider API keys

ID	Spec / area	Branch (stream)	Severity	Finding	Source
KD-026	011/024 compose	n8-013-ui-gate merge e96ff43 (candidate-introduced); fix-apikeys-argv (871dacc)	HIGH	Provider key names are passed through compose `environment:` again (`-ApiKey_OpenAI`, `-ApiKey_Anthropic`, compose.yml:192-193). podman-compose turns each into `-e NAME=value` on the `podman create` command line (visible in ps and /proc//cmdline) and in `podman inspect` Config.Env. The earlier reviews rated this MED/HIGH and fix-apikeys-argv removed it; the lines	pending branch section comple F1 (both the pre-regress digest)

ID	Spec / area	Branch (stream)	Severity	Finding	Source
				came back in the conflict-resolved n8 merge. The file's own comment at line 157 says they are deliberately not there. Only matters if an operator opts in with WORKSHOP_API_KEY_NAMES.	
KD-027	011 hosted providers	h1-providers-hardening (239a960) merged as 8c2fa19	MED	The hosted Anthropic row (billable health probe) can no longer be brought up: the probe now fails closed unless a spend hook is installed, and nothing in production code calls `answer.SetSpendHook` (callers are tests only). OpenAI and the other 28 rows probe with a GET and are unaffected. Hosted providers are off by default anyway.	small-branch H1-2 (th open); c fix obse candida

G. On-demand material generation (spec 022)

ID	Spec / area	Branch (stream)	Severity	Finding	Source repor
KD-028	022 wiring	s17-022-material (44dbdb8)	HIGH	Feature is not wired: `attachMaterialPipeline(...)` is never called, so with `-materials-enabled` the server answers a permanent 503 ('answering pipeline was not attached'); no materials route exists in the UI. Fail-safe but a learner cannot reach it.	s17-round4-2026-C1
KD-029	022 e2e	s17-022-material (44dbdb8)	HIGH	The enabled-feature e2e fails after the merge with the spec-024 gate (stub provider is forced off, 'compose'	s17-round4-2026-C2

ID	Spec / area	Branch (stream)	Severity	Finding	Source repor
				control never appears, generate count 0). The builder had not re-run it.	
KD-030	022 prompt safety	s17-022-material (44dbdb8)	MED	Prompt delimiter forgery survives `neutralise()`: <<<<< becomes < < <<<, which still contains <<<, so a passage or topic can forge a SOURCE/ request block.	s17-round4-2026- F3
KD-031	022 delete	s17-022-material (44dbdb8)	MED	Delete racing a same-topic cache-hit commit leaves a dangling material row; the learner's whole materials list then returns 404 and is unrecoverable. Reviewer rated LOW-MEDIUM.	s17-round4-2026- F4
KD-032	022 privacy	s17-022-material (44dbdb8)	LOW	Deleted topic text and session id remain physically in the SQLite WAL (secure_delete off); budget_reservation keeps the session id. 'Erasure' is logical only.	s17-round4-2026- F5
KD-033	022 auth	s17-022-material (44dbdb8)	MED	UNCONFIRMED: owner key falls back to a client-chosen X-Session header; if an unauthenticated request ever reached the handler this would be an IDOR and per-session budget bypass.	s17-round4-2026- F8

H. Learning-medium profile (spec 021)

ID	Spec / area	Branch (stream)	Severity	Finding
KD-034	021 backend source	s16-021-medium (905bf09)	MED	Every profile/recommendation endpoint answered 503 `scoring_store_absent` because the response source was the pre-013 stub.

ID	Spec / area	Branch (stream)	Severity	Finding
KD-035	021 UI	s16-021-medium (905bf09)	MED	The UI was unreachable: nothing mounted MEDIUM_ROUTES and there was no nav link.
KD-036	021 shutdown/ flags	s16-021-medium (905bf09)	MED	`medium` was missing from hookStreams, so deleting closeMedium() or addMediumFlags() survived the tests (store not closed on shutdown; -medium-* flags undefined).
KD-037	021 GET writes / streak	s16-021-medium (905bf09)	LOW	GET /api/profile/medium wrote on every call; concurrent requests could inflate the streak; verdict rows survived switching the profile off.
KD-038	021 hint and filter components	s16-021-medium (905bf09)	MED	Backlog: the in-lesson medium hint and kind filter components have no host, and MediumEngagementService is injected only by the consent component; engagement stays OFF unless WORKSHOP_MEDIUM_ENGAGEMENT=1. A tester will not see the in-context hint.

I. Deploy and registry promotion (spec 019)

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	St ca
KD-039	019 promote-web	s12-019-promotion (24c2200)	HIGH	The web promoter could not stage a real Angular bundle: it rejected	s12-round3-2026-10-21.md, F1	LI LA 're mi ch we no

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	St ca
				`<base href="/">` and `favicon.ico` as missing references, so no real promotion was possible.		sc pr re lo liv ha pe an
KD-040	019 promote-web	s12-019-promotion (24c2200)	MED	A concurrent `--stage` during `--promote` could install an index.html whose chunks (styles-/main-) were 404 (MED-HIGH in the report).	s12-round3-2026-10-21.md, F2	LI LA 'st re
KD-041	019 promote-web	s12-019-promotion (24c2200)	MED	Confinement fix relocated rather than prevented the 2026-10-21 incident class (no nounset in the build command, no size cap); TOCTOU between guard and copy;	s12-round3-2026-10-21.md, F3, F4, F5	LI LA 'no siz TO rc fai

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	St ca
				journal-append failure after the index rename exited rc 1 ('served directory unchanged') while the new bundle was live.		
KD-042	019 provers	s12-019-promotion (24c2200)	MED	prove-web-promotion failed ('INVALID W8: pattern not found') and the committed prover logs were stale, so the idempotency guard had no live mutant.	s12-round3-2026-10-21.md, F8	LI LA ch ga pr mu no DI al ch
KD-043	019 status page	s12-019-promotion (24c2200)	MED	Backlog: the 'What is live' panel `<wk-live-facts>` is not used by the status page, so the live-vs-built facts are invisible; `/api/system/provenance` is served but	completion-backlog-B-specs-018-026.md, part A section 3 item 5	OP gi fa pl fr on fa its

ID	Spec / area	Branch (stream)	Severity	Finding	Source report	St ca
				never shown. Backlog: `scripts/restart.sh` has no `--require-verdict`; a restart (what actually deploys a binary) is not gated by the deploy-gate verdict.		
KD-044	019 restart	s12-019-promotion (24c2200)	MED		completion-backlog-B-specs-018-026.md, part A section 3 item 6	OP `gr` `ve` `re` ma

J. Providers, streaming privacy, progress store (specs 011, 013, 016)

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
KD-045	011 admin audit card	h1-providers-hardening (239a960)	MED	A successful health probe (`health_check_ok`) was counted as a FAILURE on the admin audit card: every hosted provider build added one failure and one attempt.	small-branches-2026-H1-1
KD-046	011 provider health	h1-providers-hardening (239a960)	MED	Health() was vacuous for hosted providers (nil for an unprobed provider whose vendor is down),	small-branches-2026-H1-4

ID	Spec / area	Branch (stream)	Severity	Finding	Source report
				so a vendor outage showed up as a failed job instead of an immediate 'unavailable'. Reviewer rated LOW/MEDIUM.	
KD-047	011/024 streaming privacy	inv-nonascii (99fa0b7) via h1 merge 960344e	MED	A late SSE subscriber (or one that attaches after the first deltas) receives the suffix of a suppressed non-ASCII run: up to window-1 words such as 'jalapeño', 'smörgåsbord' reached the wire (10 of 10 under -race).	small-branches-2026-INV-1
KD-048	016 progress store	s2-016us1-013us1 (e2dde60)	LOW	A directory-fsync failure after the rename returned a generic 'progress store could not be read' although the new state was already published (caller saw failure, next read showed lesson complete).	small-branches-2026-S2-2

K. Completion-backlog items a tester can still hit

ID	Spec / area	Branch (stream)	Severity	Finding	S
KD-049	025 UX research wave	s15-025-ux (b344ae7)	HIGH	All eight UX flags default to false (core/ux-flags.ts:31-40) and can only be switched per browser via localStorage key `workshop-ux-flags`; with default settings none of the 025 features (staged streaming, palette, peek, comfort, momentLinks) is visible.	c s W
KD-050	025 US2 confidence prompt	s15-025-ux (b344ae7)	MED	Confidence-before-reveal (US2, T031-T041) is not built; the `practiceConfidence` flag is declared and consumed nowhere. Also the draft preview is unreachable (never passes `{tokens:true}`).	c s W
KD-051	025 reduced motion	s15-025-ux (b344ae7)	MED	The blanket `prefers-reduced-motion` block in styles.scss still sets animation/transition to .01ms for everything; the finer comfort rules (FR-010) need the flag that is off. Three LEGACY motion rows remain in the baseline.	c s W
KD-052	024 gate inputs in compose	n3 / wire-completion	MED	Compose passes WORKSHOP_ANSWER_VERDICT_DIR empty by default and mounts no verdict/threshold/benchmark inputs, so any provider other than `none` is forced off by the answering gate. Answering stays OFF (decision 2026-09-25).	c s 2
KD-053	011 US4 provider hot-switch	none (not built)	MED	A provider cannot be changed without restarting the server (FR-011/012/013 not implemented anywhere); the FAQ says 'Not yet'. Blocked on two operator decisions (mechanism and in-flight policy).	c s s r D
KD-054	018 research library	s11-018-library (5e826bb)	MED	Library has a backend for research-source links (FR-011/FR-016/US4) but no UI caller: `LibraryApi.sources()` is	c s s

ID	Spec / area	Branch (stream)	Severity	Finding	S
				never called, search hits carry no library link, no search-within-library.	
KD-055	001 chapter add	n6-001-002-complete (a5847e7)	MED	A chapter added by the documented procedure (`scripts/add-chapter.sh`) is not served until an operator restarts the server; `publish` refuses until then. Cross-reference carry-forward needs identical root_hash (SC-023).	c s s
KD-056	012/020 textbook figures	n3-020-024-complete (bf2232e)	MED	Backlog: live catalogue has `diagrams: 0` (diagram extraction never run), so the new figures page shows nothing on live data. Code path (image route + /textbooks/:id/figures) exists in the candidate.	c s s t s
KD-057	027 research traceability	umbrella specs (not in candidate)	MED	Spec 027 research/RG text still says 'no 013 task mentions' repeat-attempt/2PL work, while 013 T050-T053 built it (commit 832b020): a second parallel implementation is specified (pkg/attemptintegrity, cmd/item-fit).	u g I

Appendix A. Findings against repositories outside the candidate

Source	Severity	Finding	Status
governor-279-2026-10-22.md F1	BLOCKING	constitution branch feat/fair-concurrency-governor does not fast-forward (github main	not in candidate (constitution repo)

Source	Severity	Finding	Status
		moved to 5f8503b; conflicts in generated index/ progressive files). NO-GO as-is.	
governor-279 F2	HIGH	Governor fails OPEN when the hold record cannot be written (read-only/full state dir: maxoverlap=3 with capacity 1).	not in candidate
governor-279 F3	HIGH	Waiting is not cheap: 100 waiters collapse (27 ran, 73 rc 2); ~6.5 cores consumed; mutex age-reclaim can rob a live holder.	not in candidate
governor-279 F4-F9	MED	SIGINT/SIGQUIT ignored by governed command; post_update_hook install_governor not isolated and clobbers existing file/symlink target; CLI loops forever on missing option value; AGENTS/ QWEN/GEMINI header tables corrupted; stats needs gawk (mawk breaks); load-aware backoff degenerates to single slot under	not in candidate

Source	Severity	Finding	Status
		foreign load (p50 wait 376 s).	
umbrella-guard-027 A1	MED	Guard refuses `go test ./cmd/workshop-redact/...` (false positive; hint never mentions the WORKSHOP_ROOT prefix).	not in candidate (umbrella repo)
umbrella-guard-027 A4	MED	New guard rows raise the guard-gaps sweep from 21 to 23 HIGH findings (sweep probes only the upstream guard); stale text in guard-project-catalogue.sh:6.	not in candidate (umbrella repo)
umbrella-guard-027 B1	HIGH	Spec 027 RG-13/ RG-14 duplicate work already built by 013 T050-T053 (re-scope before Phase 3). Counted in main table as the last row.	not in candidate (umbrella repo)

Open operator decisions

#	Decision awaited	Report reference
1	Spec 021 SC-009 (95%/80% power) cannot hold as written at 30 graded answers per format. Options (a) power >=80% at 120/format (0.82 single look, 0.803 with persistence, margin 0.003), (b) null ever-shown <=5% by 120,	pending-branches-2026-10-21.md section 1 (State of the open decision); s16 commit 905bf09

#	Decision awaited	Report reference
	(c) larger effect. The 2026-10-21 operator answer in memory is option (a); the reports list it as open and the s16 round says 'SC-009 test asserts ratified bound' (905bf09). Confirm the spec text was actually reworded.	
2	Spec 011 FR-011 (M1): how an operator changes the active answering provider without restart. Recommended: authenticated admin HTTP endpoint (option B) surfaced by the settings screen. Nothing implemented.	decision-memos-011-015.md M1
3	Spec 011 FR-012 (M2): in-flight /api/ask jobs when the provider switches. Recommended: locality-aware (finish on local/LAN, cancel when leaving a hosted provider).	decision-memos-011-015.md M2
4	Spec 011 T009 (M3): confirm the audit log fails CLOSED (remote egress that cannot be audited does not start), and keep the stricter `lan` address policy.	decision-memos-011-015.md M3
5	Spec 015 FR-022 (M4): never blend per-format progress into one score (shipped) vs a blended or opt-in overall figure.	decision-memos-011-015.md M4
6	Spec 014 (M5): ratify toast numbers (maxVisible 2, coalesce 4000 ms, debounce 30000 ms, queue 50) so the [NEEDS CLARIFICATION] markers can go.	decision-memos-011-015.md M5
7		

#	Decision awaited	Report reference
	Hosted provider enablement (D-06): per-request acknowledgement, audit visibility and spend cap are in n3; the answering gate still needs the benchmark v2 independent re-label (T016) and thresholds before any hosted row can be un-forced; keys by 0600 file only; ApiKey_TypeSafe stays off-limits. Also: decide the production spend hook (see KD on hosted Anthropic).	n3-2026-10-21.md finding 8 (INFO); small-branches H1-2; completion-backlog-B section 2 item 10; n3 docs/decision-memo-answering-primitive.md
8	Spec 013 flashcard self-rating policy: damp or exclude self-rated outcomes, stop re-serving answered flashcards, persist a `self_rated` marker, and accept or fix the quadratic first-attempt query before an area can have thousands of first answerers.	n1-2026-10-21.md Conditions 1-2 (F1, F2)
9	Spec 024: whether the L4b heuristic layer should be replaced by an entailment primitive (decision memo shipped in bdc7a19); acceptance of residual 17/55 admitted fabrications while answering is OFF.	n3-rereview-2026-10-22.md section 7; docs/decision-memo-answering-primitive.md
10	Spec 021 US5 (server routing bias) is BLOCKED until 013 kind-easing is exercised; OD-2 routing, OD-3 engagement default, OD-4 retention vs erasure of the response log, OD-5 instructor visibility (needs spec 023 roles) have no operator record.	completion-backlog-B-specs-018-026.md part B (3) and (4)
11	Spec 022 materials: attach the pipeline in main.go, mount the	

#	Decision awaited	Report reference
	route and decide the compose turn-on path before materials can be shown; harness repair for the enabled e2e.	s17-round4-2026-10-21.md C1, C2
12	OR-04 (no LLM/ASR/'vision pass' over videos) vs spec 008 OCR chain T449-T465 (frame sampling + tesseract): operator ruling needed before the OCR stream starts.	completion-backlog-B-specs-018-026.md part E sub5 wiring 8
13	Spec 027: 22 NEEDS-OPERATOR ids (NO-1..NO-16, J1..J5) in spec.md, including the NO-5 attempt-integrity policy flag.	umbrella-guard-027-2026-10-21.md section B
14	Release cadence: 0.0.5 not cut; reports say cut when S11+S12+S16 are merged (already decided 2026-10-21). Waivers from 0.0.3/0.0.4 releases and the unresolved flaky ingest-textbooks test remain.	research-traceability-2026-10-21.md sections 4 and 5
15	Governor anchor 279: operator decisions on F9 (load-aware backoff degenerating to one slot) and F12 (20-agent baseline vs 6-agent cap). Local stray commit 947d402 'base' in the vasic repo from the governor review (reviewer's suggested `git reset --mixed 0748291` was denied; the next commit df25a61 removed live.jsonl).	governor-279-2026-10-22.md F9, F12, incident note
16	Rerun/accept before using any review GO as current: streams n4, n6, n7, n8, s11, s15, s18 and cat-authors are merged in the candidate but have no independent review report in	MANIFEST.tsv vs reviews directory listing

#	Decision awaited	Report reference
	the reviews directory (only the backlog audits mention them).	

Method notes and limits

- Candidate checks used `git -C <candidate> grep/show/log`; the Go, Karma and shell tests were not run for this digest.
- 'OPEN' means the code or text the report criticised is still present in the candidate tree at HEAD. 'LIKELY-FIXED-LATER' means a later commit on the merged branch tip names the finding and the changed code looks like the remedy, without a fresh independent review. 'DORMANT' means present but unreachable or off by default. 'UNKNOWN' means not determined.
- The candidate-introduced compose regression (ApiKey_ lines) is not in any review; it appears because the candidate merges both the earlier apikeys-wiring design and its supersession.
- completion-backlog A and B are audits of 2026-10-21 against int-0.0.3 and main; many of their items (practice hand-over, event source, crossref floor, FB2/DOCX/HTML extractors, scoring adapters) were completed by the n-streams and are omitted here after checking. Only items still observable in the candidate were kept.